

Fundamentals of Artificial Intelligence and Machine Learning

Complete student lesson for course code **5133C**.

Revision 2026 Semester 5 Program Elective 4 modules

Course snapshot

Scheme: 4 (L:4 T:0 P:0)

Credits: 4

Contact: 60 periods per semester

Module hours listed: 58

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.
2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Handbook method: Do not treat a topic as completed after reading its heading. For every topic card, complete the learning objectives, follow the conceptual framework, write the worked answer method, and answer the self-check questions. This creates a study record that is useful for class learning and examination preparation.

Course Dashboard

Course code

5133C

Semester

5

Credits

4

Teaching scheme

4 (L:4 T:0 P:0)

No.	Module	Official hours	Relevant CO / level
1	Fundamental Understanding of Artificial Intelligence	10	CO1
2	Python Programming for AI	20	CO2
3	Machine Learning with Python	18	CO3
4	AI with Python in Gaming	10	CO4

Prerequisites

- Basic Mathematics — Mathematics, semesters 1 and 2.
- Problem solving and Programming concepts — Problem solving and Programming, semester 2.
- Object Oriented Programming concepts — Object Oriented Programming, semester 4.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. To impart fundamental understanding of artificial intelligence and machine learning concepts.
2. To build real world artificial intelligence applications using Python.
3. To lay the foundation for the advanced courses like deep learning, neural networks, genetic algorithms.

Course Outcomes

1. CO1: Develop fundamental understanding of AI — 10 hours — Understanding.

2. CO2: Demonstrate Python program to solve the problems in AI — 20 hours — Applying.
3. CO3: Use Python in Machine Learning — 18 hours — Applying.
4. CO4: Apply AI using Python in Gaming — 10 hours — Applying.

CO-PO mapping as supplied

CO1: PO1 = 3 (Strongly mapped).

CO2: PO1 = 3 (Strongly mapped).

CO3: PO1 = 3 (Strongly mapped).

CO4: PO1 = 3 (Strongly mapped).

Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

Complete Syllabus Coverage

Official syllabus point-by-point coverage

This audit layer maps every official source block to the corresponding lesson module. The source transcript is retained verbatim as extracted from the supplied syllabus; the study guidance below is educational support and is not presented as additional official syllabus content. No official point is intentionally omitted.

Source-to-lesson coverage map

Module	Lesson topic cards	Source basis	Official point units
Module 1	4	Official Contents block	6
Module 2	4	Official Contents block	3
Module 3	5	Official Contents block	5
Module 4	5	Official Contents block	2

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
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Module	Topic code	Topic	Hours
module-1	M1.01	Artificial Intelligence and the necessity of learning AI	3
module-1	M1.02	Different types of learning	3
module-1	M1.03	Fields of AI	2
module-1	M1.04	Applications of AI	2
module-2	M2.01	Role, features and installation of Python	2
module-2	M2.02	Python basics: data types, conditionals, loops, control statements and strings	6
module-2	M2.03	List and dictionary manipulations, functions, modules and packages	6
module-2	M2.04	Object-oriented programming in Python and regular expressions	6
module-3	M3.01	Machine learning	2
module-3	M3.02	Types of machine learning, classification, regression, clustering and association	4
module-3	M3.03	Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM	6
module-3	M3.04	Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization	4
module-3	M3.05	Steps for building a classifier in Python	2
module-4	M4.01	Search algorithms	2
module-4	M4.02	Combinational search and Minimax algorithm	3
module-4	M4.03	Building bots to play games	1
module-4	M4.04	Bot to play last coin standing game	2

Module	Topic code	Topic	Hours
module-4	M4.05	Bot to play Tic Tac Toe	2

Learning Roadmap

1. Fundamental Understanding of Artificial Intelligence
CO1 · 10 hours

2. Python Programming for AI
CO2 · 20 hours

3. Machine Learning with Python
CO3 · 18 hours

4. AI with Python in Gaming
CO4 · 10 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Fundamental Understanding of Artificial Intelligence

Artificial Intelligence studies computational systems that perform tasks associated with perception, reasoning, learning, decision-making, language, or action. This module builds the vocabulary needed before programming AI and machine-learning solutions.

Hours: 10

CO1 · Bloom level: Understanding

Handbook study routine: Complete Module 1 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 1

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

Introduction to Artificial Intelligence - Necessity of Learning AI - Different types of Learning-Different fields of AI - Application of AI - AI tools and learning Models

Point-by-point study guide

➡ Official syllabus point 1: Introduction to Artificial Intelligence

Exact source point

Syllabus text: Introduction to Artificial Intelligence

How to understand and study it

This point is an official syllabus requirement: “Introduction to Artificial Intelligence”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് Introduction to Artificial Intelligence ് technical terms English-് ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Introduction to Artificial Intelligence is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Introduction to Artificial Intelligence using the official terminology retained above.
- Organise the important elements of Introduction to Artificial Intelligence into a logical sequence, classification, structure, or calculation.
- Connect Introduction to Artificial Intelligence with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on Introduction to Artificial Intelligence with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Introduction to Artificial Intelligence. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Introduction to Artificial Intelligence is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Introduction to Artificial Intelligence, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Introduction to Artificial Intelligence, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Introduction to Artificial Intelligence?
- How would you distinguish Introduction to Artificial Intelligence from a closely related idea?
- Where would an engineer or technician encounter Introduction to Artificial Intelligence, and what must be checked before using it?
- What common mistake would make an answer or operation involving Introduction to Artificial Intelligence incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Introduction to Artificial Intelligence ഫലം topic
പ്രശ്നത്തിന് ഉത്തരം exact technical term ഉപയോഗിക്കുക. ഫലം definition
principle, named parts/steps, application, limitation ഉപയോഗിച്ച്
ഉത്തരം നൽകുക. English terminology, symbols, units, diagram labels
ഉപയോഗിക്കുക. Exam answer ഫലം concept-ഫലം ഉത്തരം-ഫലം
ഉത്തരം ഉപയോഗിക്കുക.

— Official syllabus point 2: Necessity of Learning AI

Exact source point

Syllabus text: Necessity of Learning AI

How to understand and study it

This point is an official syllabus requirement: “Necessity of Learning AI”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ഏത് syllabus point ന്നതിനെയും Necessity of Learning AI ന്നതിനായി technical terms English-ൽ നൽകിയിരിക്കുന്നു. ഏത് definition ന്നതിനെയും principle ന്നതിനെയും; ഏത് term-ന്റെ function, difference, application ന്നതിനെയും നൽകിയിരിക്കുന്നു. Diagram, sequence, formula, unit ന്നതിനെയും നൽകിയിരിക്കുന്നു revision ന്നതിനെയും.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Necessity of Learning AI is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Necessity of Learning AI using the official terminology retained above.
- Organise the important elements of Necessity of Learning AI into a logical sequence, classification, structure, or calculation.
- Connect Necessity of Learning AI with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on Necessity of Learning AI with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Necessity of Learning AI. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Necessity of Learning AI is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Necessity of Learning AI, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Necessity of Learning AI, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Necessity of Learning AI?
- How would you distinguish Necessity of Learning AI from a closely related idea?
- Where would an engineer or technician encounter Necessity of Learning AI, and what must be checked before using it?
- What common mistake would make an answer or operation involving Necessity of Learning AI incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Necessity of Learning AI താഴെ പറയുന്ന വിഷയം പഠിക്കേണ്ടത് ആവശ്യമാണ്. താഴെ പറയുന്ന exact technical term ഉപയോഗിക്കേണ്ടത്. താഴെ പറയുന്ന definition ഉപയോഗിക്കേണ്ടത് principle, named parts/steps, application, limitation ഉപയോഗിക്കേണ്ടത്. English terminology, symbols, units, diagram labels ഉപയോഗിക്കേണ്ടത്. Exam answer ഉപയോഗിക്കേണ്ടത് concept-ഉപയോഗിക്കേണ്ടത്. ഉപയോഗിക്കേണ്ടത്.

— Official syllabus point 3: Different types of

Exact source point

Syllabus text: Different types of

How to understand and study it

This point is an official syllabus requirement: “Different types of”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് Different types of ് technical terms English-ൿ ൿ. ൿ definition ൿ principle ൿ; ൿ ൿ term-ൿ function, difference, application ൿ ൿ. Diagram, sequence, formula, unit ൿ ൿ ൿ revision ൿ.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Different types of is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Different types of using the official terminology retained above.
- Organise the important elements of Different types of into a logical sequence, classification, structure, or calculation.
- Connect Different types of with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on Different types of with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Different types of. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Different types of is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Different types of, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Different types of, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Different types of?
- How would you distinguish Different types of from a closely related idea?
- Where would an engineer or technician encounter Different types of, and what must be checked before using it?
- What common mistake would make an answer or operation involving Different types of incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Different types of താഴെ topic നൽകുന്നതിനുള്ള താഴെ exact technical term നൽകുന്നതിനുള്ള. താഴെ definition നൽകുന്നതിനുള്ള principle, named parts/steps, application, limitation നൽകുന്നതിനുള്ള താഴെ താഴെ താഴെ. English terminology, symbols, units, diagram labels നൽകുന്നതിനുള്ള താഴെ. Exam answer നൽകുന്നതിനുള്ള concept-താഴെ താഴെ-താഴെ താഴെ താഴെ താഴെ താഴെ.

– Official syllabus point 4: Learning-Different fields of AI

Exact source point

Syllabus text: Learning-Different fields of AI

How to understand and study it

This point is an official syllabus requirement: “Learning-Different fields of AI”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് Learning-Different fields of AI ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Learning-Different fields of AI is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Learning-Different fields of AI using the official terminology retained above.
- Organise the important elements of Learning-Different fields of AI into a logical sequence, classification, structure, or calculation.
- Connect Learning-Different fields of AI with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on Learning-Different fields of AI with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Learning-Different fields of AI. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Learning-Different fields of AI is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Learning-Different fields of AI, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Learning-Different fields of AI, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Learning-Different fields of AI?
- How would you distinguish Learning-Different fields of AI from a closely related idea?
- Where would an engineer or technician encounter Learning-Different fields of AI, and what must be checked before using it?
- What common mistake would make an answer or operation involving Learning-Different fields of AI incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Learning-Different fields of AI താഴെ topic

താഴെ പറയുന്നവയെക്കുറിച്ച് താഴെ exact technical term ഉപയോഗിക്കുക. definition
principle, named parts/steps, application, limitation ഉപയോഗിച്ച്
താഴെ പറയുന്നവയെക്കുറിച്ച്. English terminology, symbols, units, diagram labels
ഉപയോഗിച്ച് ഉപയോഗിക്കുക. Exam answer ഉപയോഗിച്ച് concept-ഉപയോഗിച്ച് ഉപയോഗിക്കുക-ഉപയോഗിച്ച്
ഉപയോഗിക്കുക ഉപയോഗിക്കുക.

— Official syllabus point 5: Application of AI

Exact source point

Syllabus text: Application of AI

How to understand and study it

This point is an official syllabus requirement: “Application of AI”. Connect each named application to the technical concept: identify the requirement, the component or method used, and the result obtained. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് Application of AI ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Application of AI is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Application of AI using the official terminology retained above.
- Organise the important elements of Application of AI into a logical sequence, classification, structure, or calculation.
- Connect Application of AI with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on Application of AI with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Application of AI. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Application of AI is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Application of AI, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Application of AI, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Application of AI?
- How would you distinguish Application of AI from a closely related idea?
- Where would an engineer or technician encounter Application of AI, and what must be checked before using it?
- What common mistake would make an answer or operation involving Application of AI incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Application of AI താഴെ topic നൽകുന്നതിനുള്ള ഉത്തരം
exact technical term നൽകുന്നതിനുള്ള. ഉത്തരം definition നൽകുന്നതിനുള്ള principle,
named parts/steps, application, limitation നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള.
English terminology, symbols, units, diagram labels നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള.
Exam answer നൽകുന്നതിനുള്ള concept-ഉത്തരം നൽകുന്നതിനുള്ള-ഉത്തരം നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള.

— Official syllabus point 6: AI tools and learning Models

Exact source point

Syllabus text: AI tools and learning Models

How to understand and study it

This point is an official syllabus requirement: “AI tools and learning Models”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് AI tools, learning Models
് technical terms English-് ്. ് definition ്
principle ്; ് term-് function, difference, application
്. Diagram, sequence, formula, unit ്
് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

AI tools and learning Models is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope AI tools and learning Models using the official terminology retained above.
- Organise the important elements of AI tools and learning Models into a logical sequence, classification, structure, or calculation.
- Connect AI tools and learning Models with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on AI tools and learning Models with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading AI tools and learning Models. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of AI tools and learning Models is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on AI tools and learning Models, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is AI tools and learning Models, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining AI tools and learning Models?
- How would you distinguish AI tools and learning Models from a closely related idea?
- Where would an engineer or technician encounter AI tools and learning Models, and what must be checked before using it?
- What common mistake would make an answer or operation involving AI tools and learning Models incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: AI tools and learning Models താഴെ topic

താഴെ പറയുന്നവയെക്കുറിച്ച് താഴെ exact technical term താഴെ പറയുന്നവയെക്കുറിച്ച്. താഴെ definition താഴെ പറയുന്നവയെക്കുറിച്ച് principle, named parts/steps, application, limitation താഴെ പറയുന്നവയെക്കുറിച്ച്. English terminology, symbols, units, diagram labels താഴെ പറയുന്നവയെക്കുറിച്ച്. Exam answer താഴെ പറയുന്നവയെക്കുറിച്ച് concept-താഴെ പറയുന്നവയെക്കുറിച്ച് താഴെ പറയുന്നവയെക്കുറിച്ച്.

Learning goals

- Define AI and explain why learning AI is important.
- Classify learning approaches at an introductory level.
- Identify fields, applications, tools, and learning models.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

Concept and explanation

Artificial Intelligence is the design of systems that perform tasks requiring capabilities such as perception, reasoning, planning, language understanding, learning, or decision-making. Learning AI is necessary because data, automation, intelligent assistance, robotics, recommendation, and decision-support are present across engineering and society. A responsible AI learner also studies data quality, bias, privacy, safety, and limits of automated decisions.

Malayalam support: AI perception, reasoning, planning, language, learning, decision-making capabilities computer system-ൽ ഉപയോഗിക്കുന്നു. AI ഉപയോഗിക്കുമ്പോൾ data quality, bias, privacy, safety പരിശോധിക്കേണ്ടതാണ്.

Study points

- Define AI using task capability rather than a single product name.
- State engineering and societal reasons for learning AI.
- Mention limitations and responsible-use concerns.

Engineering / workplace application: AI is used in predictive maintenance, quality inspection, navigation, natural-language interfaces, and resource optimisation.

Illustrative example: A maintenance model can learn from sensor history to flag likely equipment failure, but a technician still validates the recommendation.

Common mistake: Assuming a system is intelligent merely because it uses a large dataset confuses automation with validated intelligence.

Handbook treatment

M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours) using the official terminology retained above.
- Organise the important elements of M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours)?
- How would you distinguish M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.01 — Artificial Intelligence and the necessity of learning AI (3 hours) താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term ഉപയോഗിക്കുക. definition, principle, named parts/steps, application, limitation എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer concept-ബ്ലോക്ക് രീതിയിൽ ഉപയോഗിക്കുക.

– M1.02 — Different types of learning (3 hours)

Concept and explanation

At an introductory level, supervised learning uses labelled examples to learn a mapping for prediction or classification. Unsupervised learning searches for structure in unlabelled data, such as clusters or associations. Reinforcement learning learns actions from interaction, rewards, and penalties. The choice of learning type depends on available data, target task, feedback, and evaluation method.

Malayalam support: Labelled data ശുപരിവേദിതം supervised learning; label ലേബൽ data-ശുപരിവേദിതം structure ശുപരിവേദിതം unsupervised; reward/penalty പ്രതിഫലം/പ്രതിരോധം action പ്രവർത്തനം reinforcement learning. Data and task ശുപരിവേദിതം type ശുപരിവേദിതം.

Study points

- Compare input labels, feedback, and output for the major learning types.
- Give one engineering example for each type.
- Relate evaluation to the learning objective.

Engineering / workplace application: Fault classification can be supervised, customer segmentation can be unsupervised, and robot navigation can use reinforcement learning.

Illustrative example: A labelled set of normal and failed motor readings can support supervised classification, while unlabelled readings can be grouped to explore operating states.

Common mistake: Unsupervised clustering does not automatically provide meaningful human categories; interpretation is required.

Handbook treatment

M1.02 — Different types of learning (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.02 — Different types of learning (3 hours) using the official terminology retained above.
- Organise the important elements of M1.02 — Different types of learning (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.02 — Different types of learning (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on M1.02 — Different types of learning (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.02 — Different types of learning (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.02 — Different types of learning (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.02 — Different types of learning (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.02 — Different types of learning (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.02 — Different types of learning (3 hours)?
- How would you distinguish M1.02 — Different types of learning (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.02 — Different types of learning (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.02 — Different types of learning (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.02 — Different types of learning (3 hours) താഴെ പറയുന്ന വിഷയം പഠിക്കുക. താഴെ പറയുന്ന exact technical term ഉപയോഗിക്കുക. definition principle, named parts/steps, application, limitation ഉപയോഗിച്ച് വിശദീകരിക്കുക. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer concept-പദം ഉപയോഗിച്ച് വിശദീകരിക്കുക.

Concept and explanation

AI includes fields such as machine learning, natural-language processing, computer vision, robotics, expert systems, search and planning, speech processing, knowledge representation, and game AI. Fields often overlap: a robot may use vision, planning, learning, and control together. The field should be selected from the problem capability required, not from a fashionable label.

Malayalam support: Machine learning, NLP, computer vision, robotics, expert systems, search/planning, speech, knowledge representation, game AI ഫീൽഡുകൾ AI fields ഫീൽഡുകൾ. ഫീൽഡ് application ഫീൽഡ് fields ഫീൽഡുകൾ.

Study points

- List major fields and the capability each addresses.
- Explain that fields can be combined in one application.
- Distinguish AI field from programming language or tool.

Engineering / workplace application: Industrial inspection may combine computer vision, machine learning, and robotic control.

Illustrative example: A voice-controlled robot combines speech processing, language understanding, planning, and embedded control.

Common mistake: Python or a library is not itself a field of AI; it is a means of implementing an AI solution.

Handbook treatment

M1.03 — Fields of AI (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.03 — Fields of AI (2 hours) using the official terminology retained above.
- Organise the important elements of M1.03 — Fields of AI (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.03 — Fields of AI (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on M1.03 — Fields of AI (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.03 — Fields of AI (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.03 — Fields of AI (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.03 — Fields of AI (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.03 — Fields of AI (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.03 — Fields of AI (2 hours)?
- How would you distinguish M1.03 — Fields of AI (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.03 — Fields of AI (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.03 — Fields of AI (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.03 — Fields of AI (2 hours) താഴെ വിഷയം പഠിക്കുന്നതിനായി ഉപയോഗിക്കേണ്ടതാണ് exact technical term ഉപയോഗിക്കേണ്ടതാണ്. താഴെ definition ഉപയോഗിക്കേണ്ടതാണ് principle, named parts/steps, application, limitation ഉപയോഗിക്കേണ്ടതാണ്. English terminology, symbols, units, diagram labels ഉപയോഗിക്കേണ്ടതാണ്. Exam answer ഉപയോഗിക്കേണ്ടതാണ് concept-ഉപയോഗിക്കേണ്ടതാണ് ഉപയോഗിക്കേണ്ടതാണ്.

Concept and explanation

Applications include image and speech recognition, recommendation, medical and industrial decision support, autonomous systems, fraud detection, predictive maintenance, chat interfaces, games, and optimisation. Each application has a task, data, model or rule system, evaluation metric, deployment environment, and risk profile. A useful application must be validated against real requirements and monitored after deployment.

Malayalam support: AI applications image/speech recognition, recommendation, predictive maintenance, fraud detection, games, optimisation. application- task, data, model, evaluation, deployment, risk.

Study points

- Describe task, data, model, evaluation, and deployment in an AI application.
- State one benefit and one risk of an application.
- Connect application choice to engineering needs.

Engineering / workplace application: AI can assist engineers by detecting anomalies, prioritising inspections, or optimising schedules.

Illustrative example: A visual inspection model should be tested on lighting, product variation, and defect cases that represent the production line.

Common mistake: A high accuracy score on an unrepresentative dataset does not guarantee safe real-world performance.

Handbook treatment

M1.04 — Applications of AI (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.04 — Applications of AI (2 hours) using the official terminology retained above.
- Organise the important elements of M1.04 — Applications of AI (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.04 — Applications of AI (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Fundamental Understanding of Artificial Intelligence.
- Write an examination answer on M1.04 — Applications of AI (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.04 — Applications of AI (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.04 — Applications of AI (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.04 — Applications of AI (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.04 — Applications of AI (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.04 — Applications of AI (2 hours)?
- How would you distinguish M1.04 — Applications of AI (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.04 — Applications of AI (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.04 — Applications of AI (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.04 — Applications of AI (2 hours) താഴെ topic ഉൾപ്പെടെയുള്ളവയെക്കുറിച്ച് exact technical term ഉൾപ്പെടെയുള്ളവയെക്കുറിച്ച്. definition ഉൾപ്പെടെയുള്ളവ principle, named parts/steps, application, limitation ഉൾപ്പെടെയുള്ളവ ഉൾപ്പെടെയുള്ളവ. English terminology, symbols, units, diagram labels ഉൾപ്പെടെയുള്ളവ ഉൾപ്പെടെയുള്ളവ. Exam answer ഉൾപ്പെടെയുള്ളവ concept-ഉൾപ്പെടെയുള്ളവ ഉൾപ്പെടെയുള്ളവ ഉൾപ്പെടെയുള്ളവ.

Module summary: AI is a broad problem-solving field; learning type, field, application, data, and evaluation must be connected.

OFFICIAL MODULE 2

Python Programming for AI

Python provides readable syntax, a large ecosystem, and rapid experimentation for AI programs. This module revises core programming and adds reusable structures needed for later data and learning tasks.

Hours: 20

CO2 · Bloom level: Applying

Handbook study routine: Complete Module 2 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 2

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

[View extracted official source transcript](#)

Role of python in AI-Features of python-Installing python-Python:- Basics Data Types, Conditional Statements, Looping, Control Statements, String, List And Dictionary Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling

Point-by-point study guide

Exact source point

Syllabus text: Role of python in AI-Features of python-Installing python-Python:- Basics Data Types

How to understand and study it

This point is an official syllabus requirement: “Role of python in AI-Features of python-Installing python-Python:- Basics Data Types”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: syllabus point റോൾ ഓഫ് പൈതൻ ഇൻ ഐ-
 Features of python-Installing python-Python:- Basics Data Types ടെക്സ്റ്റ് ടെക്സ്റ്റ്
 terms English-റോൾ ഓഫ് പൈതൻ. പൈതൻ definition പ്രിൻസിപ്പിൾ
 പ്രിൻസിപ്പിൾ; ടെക്സ്റ്റ് ടെക്സ്റ്റ് term-റോൾ function, difference, application ടെക്സ്റ്റ്
 ടെക്സ്റ്റ് ടെക്സ്റ്റ്. Diagram, sequence, formula, unit ടെക്സ്റ്റ് ടെക്സ്റ്റ് ടെക്സ്റ്റ്
 ടെക്സ്റ്റ് revision ടെക്സ്റ്റ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Role of python in AI-Features of python-Installing python-Python:- Basics Data Types is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Role of python in AI-Features of python-Installing python-Python:- Basics Data Types using the official terminology retained above.
- Organise the important elements of Role of python in AI-Features of python-Installing python-Python:- Basics Data Types into a logical sequence, classification, structure, or calculation.
- Connect Role of python in AI-Features of python-Installing python-Python:- Basics Data Types with one engineering decision, operation, measurement, or safety requirement in the context of Python Programming for AI.
- Write an examination answer on Role of python in AI-Features of python-Installing python-Python:- Basics Data Types with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Role of python in AI-Features of python-Installing python-Python:- Basics Data Types. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Role of python in AI-Features of python-Installing python-Python:- Basics Data Types is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Role of python in AI-Features of python-Installing python-Python:- Basics Data Types, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Role of python in AI-Features of python-Installing python-Python:- Basics Data Types, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Role of python in AI-Features of python-Installing python-Python:- Basics Data Types?
- How would you distinguish Role of python in AI-Features of python-Installing python-Python:- Basics Data Types from a closely related idea?
- Where would an engineer or technician encounter Role of python in AI-Features of python-Installing python-Python:- Basics Data Types, and what must be checked before using it?
- What common mistake would make an answer or operation involving Role of python in AI-Features of python-Installing python-Python:- Basics Data Types incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Role of python in AI-Features of python-Installing python-Python:- Basics Data Types താഴെ topic നൽകുന്നതിനുള്ള ഉത്തരം exact technical term നൽകുന്നതിനുള്ള. താഴെ definition നൽകുന്നതിനുള്ള principle, named parts/steps, application, limitation നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള. English terminology, symbols, units, diagram labels നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള. Exam answer നൽകുന്നതിനുള്ള concept-നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള.

– Official syllabus point 2: Conditional Statements, Looping, Control Statements, String, List And Dictionary

Exact source point

Syllabus text: Conditional Statements, Looping, Control Statements, String, List And Dictionary

How to understand and study it

This point is an official syllabus requirement: “Conditional Statements, Looping, Control Statements, String, List And Dictionary”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് Conditional Statements, Looping, Control Statements, String ് technical terms English-്
്. ് definition ് principle ്; ് term-
് function, difference, application ്. Diagram,
sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Conditional Statements, Looping, Control Statements, String, List And Dictionary should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

Learning objectives

- Define and scope Conditional Statements, Looping, Control Statements, String, List And Dictionary using the official terminology retained above.
- Organise the important elements of Conditional Statements, Looping, Control Statements, String, List And Dictionary into a logical sequence, classification, structure, or calculation.
- Connect Conditional Statements, Looping, Control Statements, String, List And Dictionary with one engineering decision, operation, measurement, or safety requirement in the context of Python Programming for AI.
- Write an examination answer on Conditional Statements, Looping, Control Statements, String, List And Dictionary with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Conditional Statements, Looping, Control Statements, String, List And Dictionary. It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

Engineering application lens

Engineering use of Conditional Statements, Looping, Control Statements, String, List And Dictionary depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

Worked answer method

Given: the quantity to be sensed, the operating environment, and the required decision or control action.

Required: a suitable measurement chain or an explanation of how the instrument operates.

Method: map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.

Answer: state the measured quantity, principle, important characteristics, application, and limitation.

Exam answer blueprint

For a short-answer question on Conditional Statements, Looping, Control Statements, String, List And Dictionary, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Conditional Statements, Looping, Control Statements, String, List And Dictionary, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Conditional Statements, Looping, Control Statements, String, List And Dictionary?
- How would you distinguish Conditional Statements, Looping, Control Statements, String, List And Dictionary from a closely related idea?
- Where would an engineer or technician encounter Conditional Statements, Looping, Control Statements, String, List And Dictionary, and what must be checked before using it?
- What common mistake would make an answer or operation involving Conditional Statements, Looping, Control Statements, String, List And Dictionary incorrect?

Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

Malayalam support: Conditional Statements, Looping, Control

Statements, String, List And Dictionary **topic** **topic** **topic**
exact technical term **topic**. **topic** definition **topic** principle,
named parts/steps, application, limitation **topic** **topic** **topic**.
English terminology, symbols, units, diagram labels **topic** **topic**.
Exam answer **topic** concept-**topic** **topic**-**topic** **topic** **topic**.

– Official syllabus point 3: Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling

Exact source point

Syllabus text: Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling

How to understand and study it

This point is an official syllabus requirement: “Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling”. Study the input, logic or algorithm, output, and one test condition. Write the steps in order and identify the common error that changes the result. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Practise one small input-output example and test at least one boundary or fault condition.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് Manipulations, Python Functions, Modules, Packages ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling using the official terminology retained above.
- Organise the important elements of Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling into a logical sequence, classification, structure, or calculation.
- Connect Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling with one engineering decision, operation, measurement, or safety requirement in the context of Python Programming for AI.
- Write an examination answer on Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a

signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling?
- How would you distinguish Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling from a closely related idea?
- Where would an engineer or technician encounter Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling, and what must be checked before using it?
- What common mistake would make an answer or operation involving Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

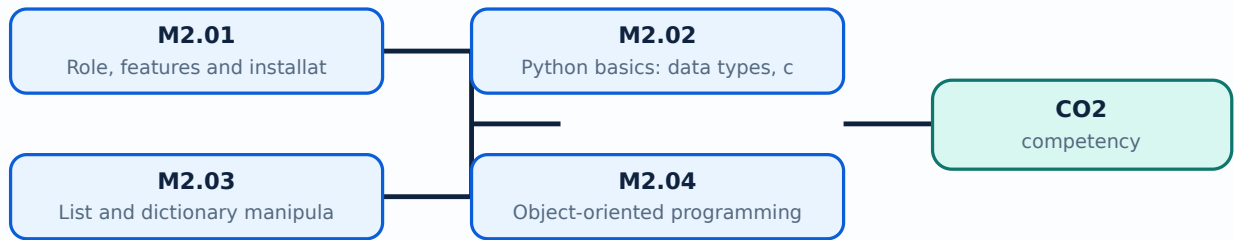
Malayalam support: Manipulations, Python Functions, Modules And Packages, Object Oriented Programming in Python, Regular Expressions, Exception Handling താഴെ topic നൽകുന്നതിനായി താഴെ exact technical term നൽകുന്നതിനായി. താഴെ definition നൽകുന്നതിനായി principle, named parts/steps, application, limitation നൽകുന്നതിനായി താഴെ താഴെ. English terminology, symbols, units, diagram labels നൽകുന്നതിനായി. Exam answer നൽകുന്നതിനായി concept-താഴെ താഴെ-താഴെ താഴെ താഴെ.

Learning goals

- Explain Python role, features, and installation for AI work.
- Write programs using data types, conditions, loops, strings, lists, dictionaries, functions, modules, and packages.
- Use object-oriented programming, regular expressions, and exception handling in Python.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

– M2.01 — Role, features and installation of Python (2 hours)

Concept and explanation

Python is used in AI because its syntax supports rapid experimentation and its ecosystem includes numerical, data, visualisation, machine-learning, and automation libraries. Important features include interpreted execution, dynamic typing, indentation-based blocks, portability, and extensive modules. Installation requires a supported Python interpreter and a controlled package environment; version and dependency records improve reproducibility.

Malayalam support: AI-ൽ Python popular ഏതെങ്കിലും readable syntax, libraries, rapid experimentation എന്നിവയ്ക്ക് അനുകൂലമാണ്. Interpreter version and package environment എന്നിവയ്ക്ക് അനുകൂലമാണ് reproducibility-യ്ക്ക് അനുകൂലമാണ്.

Study points

- State Python features relevant to AI.
- Explain interpreter, package, and environment concepts.
- Record version and dependencies for reproducibility.

Engineering / workplace application: A project may use a virtual environment to isolate numerical and machine-learning package versions.

Illustrative example: A project README can record the Python version and required packages before a classifier is shared.

Common mistake: Installing packages globally without tracking versions can make a working program difficult to reproduce.

Handbook treatment

M2.01 — Role, features and installation of Python (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.01 — Role, features and installation of Python (2 hours) using the official terminology retained above.
- Organise the important elements of M2.01 — Role, features and installation of Python (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.01 — Role, features and installation of Python (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Python Programming for AI.
- Write an examination answer on M2.01 — Role, features and installation of Python (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.01 — Role, features and installation of Python (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.01 — Role, features and installation of Python (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.01 — Role, features and installation of Python (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.01 — Role, features and installation of Python (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.01 — Role, features and installation of Python (2 hours)?
- How would you distinguish M2.01 — Role, features and installation of Python (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.01 — Role, features and installation of Python (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.01 — Role, features and installation of Python (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.01 — Role, features and installation of Python (2 hours)

topic exact technical term definition principle, named parts/steps, application, limitation English terminology, symbols, units, diagram labels Exam answer concept- -

– M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours)

Concept and explanation

Python basics include numeric, Boolean, string, list, tuple, set, dictionary, and None values; variables refer to objects and expressions produce values. Conditional statements select paths, loops repeat work, and break/continue control iteration. Strings are immutable sequences supporting indexing, slicing, methods, formatting, and conversion. Correct type conversion and boundary testing are essential when preparing input for an AI program.

Malayalam support: Python data types, if/elif/else, loops, break/continue, strings
 `AI code-` . Input string numeric conversion and validation .

Study points

- Trace a program with variables, conditions, and loops.
- Use string operations and safe type conversion.
- Test empty, boundary, and invalid inputs.

Engineering / workplace application: Data-cleaning scripts use loops and string processing before data enters a model.

Illustrative example: A program can read a temperature string, convert it to float, reject invalid input, and classify the operating condition.

Common mistake: Mixing strings and numbers without explicit conversion creates errors or incorrect comparisons.

Handbook treatment

M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours) should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

Learning objectives

- Define and scope M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours) using the official terminology retained above.
- Organise the important elements of M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Python Programming for AI.
- Write an examination answer on M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours). It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

Engineering application lens

Engineering use of M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours) depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

Worked answer method

Given: the quantity to be sensed, the operating environment, and the required decision or control action.

Required: a suitable measurement chain or an explanation of how the instrument operates.

Method: map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.

Answer: state the measured quantity, principle, important characteristics, application, and limitation.

Exam answer blueprint

For a short-answer question on M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours)?
- How would you distinguish M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours) incorrect?

Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

Malayalam support: M2.02 — Python basics: data types, conditionals, loops, control statements and strings (6 hours) **topic** **concept** **exact** **technical term** **definition** **principle**, named parts/steps, application, limitation **terminology**, symbols, units, diagram labels **Exam answer** **concept**-**terminology**-**units** **terminology**.

– M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours)

Concept and explanation

Lists store ordered mutable data; dictionaries store key-value records; functions package reusable logic; modules and packages organise code and dependencies. AI programs commonly represent samples as lists, records as dictionaries, and preprocessing or prediction steps as functions. Clear interfaces, return values, descriptive names, and small tests make experimentation reliable.

Malayalam support: List ordered data ഏകദേശം; dictionary key-value record; function reusable logic; module/package code organisation ഏകദേശം. AI program-ൽ data preparation and prediction functions ഉപയോഗിക്കുന്നു.

Study points

- Create and manipulate lists and dictionaries.
- Define functions with parameters and return values.
- Import modules and explain why packages and dependency management matter.

Engineering / workplace application: Feature extraction, data loading, and evaluation are often implemented as separate functions or modules.

Illustrative example: A `prepare_record()` function can return a dictionary of validated features that another function passes to a classifier.

Common mistake: Putting all model, input, and output code in one long script makes errors harder to isolate.

Handbook treatment

M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours) using the official terminology retained above.
- Organise the important elements of M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Python Programming for AI.
- Write an examination answer on M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours)?
- How would you distinguish M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.03 — List and dictionary manipulations, functions, modules and packages (6 hours) താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term ഉപയോഗിക്കുക. definition ഉൾപ്പെടെ principle, named parts/steps, application, limitation തുടങ്ങിയവ ഉൾപ്പെടെ. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer concept-ബ്ലോക്ക് ഉപയോഗിച്ച്-പോലെ ഉത്തരം നൽകേണ്ടതാണ്.

— M2.04 — Object-oriented programming in Python and regular expressions (6 hours)

Concept and explanation

Object-oriented programming groups data and behaviour into classes and objects. Encapsulation, constructors, methods, inheritance, and polymorphism can organise reusable AI components, although simple functions may be sufficient for small tasks. Regular expressions describe text patterns for searching, validation, extraction, and substitution. Exception handling with try/except prevents expected input or file errors from crashing a complete application and should not hide unknown failures.

Malayalam support: Class/object data and behaviour ക്ലാസ്സുകൾ/ഓബ്ജക്റ്റുകൾ organize ചെയ്യാം. Regular expression text pattern search/validation-രീതിയിൽ പാറ്റേണുകൾ തിരയുക. try/except expected errors handle ചെയ്യുക; അപ്രതീക്ഷിത errors-കൾ ഒന്നും മറയ്ക്കരുത്.

Study points

- Define a class with attributes and methods.
- Use a regular expression for a controlled extraction or validation task.
- Handle expected exceptions and report useful messages.

Engineering / workplace application: An AI data pipeline may use a class for dataset metadata and regular expressions to extract fields from text.

Illustrative example: A parser can catch a missing file error with a clear message while allowing an unexpected programming error to be detected during testing.

Common mistake: Catching every exception and doing nothing can hide corrupted data or programming defects.

Handbook treatment

M2.04 — Object-oriented programming in Python and regular expressions (6 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M2.04 — Object-oriented programming in Python and regular expressions (6 hours) using the official terminology retained above.
- Organise the important elements of M2.04 — Object-oriented programming in Python and regular expressions (6 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.04 — Object-oriented programming in Python and regular expressions (6 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Python Programming for AI.
- Write an examination answer on M2.04 — Object-oriented programming in Python and regular expressions (6 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.04 — Object-oriented programming in Python and regular expressions (6 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M2.04 — Object-oriented programming in Python and regular expressions (6 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M2.04 — Object-oriented programming in Python and regular expressions (6 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.04 — Object-oriented programming in Python and regular expressions (6 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.04 — Object-oriented programming in Python and regular expressions (6 hours)?
- How would you distinguish M2.04 — Object-oriented programming in Python and regular expressions (6 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.04 — Object-oriented programming in Python and regular expressions (6 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.04 — Object-oriented programming in Python and regular expressions (6 hours) incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M2.04 — Object-oriented programming in Python and regular expressions (6 hours) താഴെ topic നൽകുന്നതിനായി ഉപയോഗിക്കേണ്ട exact technical term നൽകുന്നതിനായി. താഴെ definition നൽകുന്നതിനായി principle, named parts/steps, application, limitation നൽകുന്നതിനായി ഉപയോഗിക്കേണ്ട. English terminology, symbols, units, diagram labels നൽകുന്നതിനായി ഉപയോഗിക്കേണ്ട. Exam answer നൽകുന്നതിനായി concept-നൽകുന്നതിനായി ഉപയോഗിക്കേണ്ട.

Module summary: Python fundamentals become useful AI skills when code is modular, validated, reproducible, and connected to data.

OFFICIAL MODULE 3

Machine Learning with Python

Machine learning turns data into a predictive or descriptive model. This module covers learning types, common algorithms, preprocessing, and a repeatable classifier-building sequence.

Hours: 18

CO3 · Bloom level: Applying

Handbook study routine: Complete Module 3 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 3

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy, Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization, Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes,Support Vector Machines (SVM),Decision Tree classifier, Random Forest.

Point-by-point study guide

– **Official syllabus point 1: MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support**

Exact source point

Syllabus text: MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support

How to understand and study it

This point is an official syllabus requirement: “MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് MachineLearning - Types of Machine learning-Supervised, Unsupervised Learning -Classification, Regression, Linear Regression ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support using the official terminology retained above.
- Organise the important elements of MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support into a logical sequence, classification, structure, or calculation.
- Connect MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K

Means, Regression, Support is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.

Required: construction, working, selection, performance, or diagnosis as requested.

Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.

Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support?
- How would you distinguish MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support from a closely related idea?
- Where would an engineer or technician encounter MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support, and what must be checked before using it?

- What common mistake would make an answer or operation involving MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: MachineLearning - Types of Machine learning-Supervised and Unsupervised Learning -Classification and Regression, Linear Regression, KNN, K Means, Regression, Support
 താഴെ പറയുന്ന വിഷയം
 താഴെ പറയുന്ന വിഷയം exact technical term
 താഴെ പറയുന്ന വിഷയം definition
 താഴെ പറയുന്ന വിഷയം principle, named parts/steps, application, limitation
 താഴെ പറയുന്ന വിഷയം
 താഴെ പറയുന്ന വിഷയം. English terminology, symbols, units, diagram labels
 താഴെ പറയുന്ന വിഷയം. Exam answer
 താഴെ പറയുന്ന വിഷയം concept-
 താഴെ പറയുന്ന വിഷയം
 താഴെ പറയുന്ന വിഷയം.

– Official syllabus point 2: Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy

Exact source point

Syllabus text: Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy

How to understand and study it

This point is an official syllabus requirement: “Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy”. Study the sequence from input or initial condition to operation and final output. Note the role of each named stage and the cause-and-effect relationship. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy using the official terminology retained above.
- Organise the important elements of Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy into a logical sequence, classification, structure, or calculation.
- Connect Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.

Required: construction, working, selection, performance, or diagnosis as requested.

Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.

Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy?
- How would you distinguish Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy from a closely related idea?
- Where would an engineer or technician encounter Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy, and what must be checked before using it?
- What common mistake would make an answer or operation involving Vector Machines (SVM), Data Preparation - Preprocessing the Data using numpy incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: Vector Machines (SVM), Data Preparation -
Preprocessing the Data using numpy $\square\square\square\square$ topic $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ exact
technical term $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ definition $\square\square\square\square\square\square\square\square$ principle,
named parts/steps, application, limitation $\square\square\square\square\square$ $\square\square\square\square\square\square\square$ $\square\square\square\square\square\square\square$.
English terminology, symbols, units, diagram labels $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$.
Exam answer $\square\square\square\square\square\square\square\square$ concept- $\square\square\square\square$ $\square\square\square\square$ - $\square\square\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square\square\square$.

Exam

How

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Handbook treatment

Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization using the official terminology retained above.
- Organise the important elements of Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization into a logical sequence, classification, structure, or calculation.
- Connect Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization?
- How would you distinguish Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization from a closely related idea?
- Where would an engineer or technician encounter Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization, and what must be checked before using it?
- What common mistake would make an answer or operation involving Techniques for Data Preprocessing- Binarization-Mean Removal, Scaling, Normalization incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Techniques for Data Preprocessing- Binarization- Mean Removal, Scaling, Normalization $\square\square\square\square$ topic $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ exact technical term $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ definition $\square\square\square\square\square\square\square\square$ principle, named parts/steps, application, limitation $\square\square\square\square\square$ $\square\square\square\square\square\square\square$ $\square\square\square\square\square\square$. English terminology, symbols, units, diagram labels $\square\square\square\square\square$ $\square\square\square\square\square\square\square$. Exam answer $\square\square\square\square\square\square\square\square$ concept- $\square\square\square\square$ $\square\square\square\square$ - $\square\square\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square\square\square$.

– Official syllabus point 4: Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier

Exact source point

Syllabus text: Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes,Support Vector Machines (SVM),Decision Tree classifier

How to understand and study it

This point is an official syllabus requirement: “Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: syllabus point ക്ലാസ്സിഫിക്കേഷൻ Classification, Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier ക്ലാസ്സ് technical terms English-ക്ലാസ്സിഫിക്കേഷൻ ക്ലാസ്സിഫിക്കേഷൻ. ക്ലാസ്സിഫിക്കേഷൻ definition ക്ലാസ്സിഫിക്കേഷൻ principle ക്ലാസ്സിഫിക്കേഷൻ; ക്ലാസ്സ് ക്ലാസ്സ് term-ക്ലാസ്സ് function, difference, application ക്ലാസ്സിഫിക്കേഷൻ ക്ലാസ്സിഫിക്കേഷൻ ക്ലാസ്സിഫിക്കേഷൻ. Diagram, sequence, formula, unit ക്ലാസ്സിഫിക്കേഷൻ ക്ലാസ്സിഫിക്കേഷൻ ക്ലാസ്സിഫിക്കേഷൻ revision ക്ലാസ്സിഫിക്കേഷൻ.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier using the official terminology retained above.
- Organise the important elements of Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier into a logical sequence, classification, structure, or calculation.
- Connect Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier is selected by matching the required duty with

capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.

Required: construction, working, selection, performance, or diagnosis as requested.

Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.

Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier?
- How would you distinguish Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier from a closely related idea?
- Where would an engineer or technician encounter Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier, and what must be checked before using it?

- What common mistake would make an answer or operation involving Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: Classification and Regression, Steps for Building a Classifier in Python, Building classifier in python - Naïve Bayes, Support Vector Machines (SVM), Decision Tree classifier

topic

exact technical term . definition principle, named parts/steps, application, limitation English terminology, symbols, units, diagram labels Exam answer concept-

– Official syllabus point 5: Random Forest.

Exact source point

Syllabus text: Random Forest.

How to understand and study it

This point is an official syllabus requirement: “Random Forest.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് Random Forest. ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Random Forest. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Random Forest. using the official terminology retained above.
- Organise the important elements of Random Forest. into a logical sequence, classification, structure, or calculation.
- Connect Random Forest. with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on Random Forest. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Random Forest.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Random Forest. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Random Forest., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Random Forest., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Random Forest.?
- How would you distinguish Random Forest. from a closely related idea?
- Where would an engineer or technician encounter Random Forest., and what must be checked before using it?
- What common mistake would make an answer or operation involving Random Forest. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

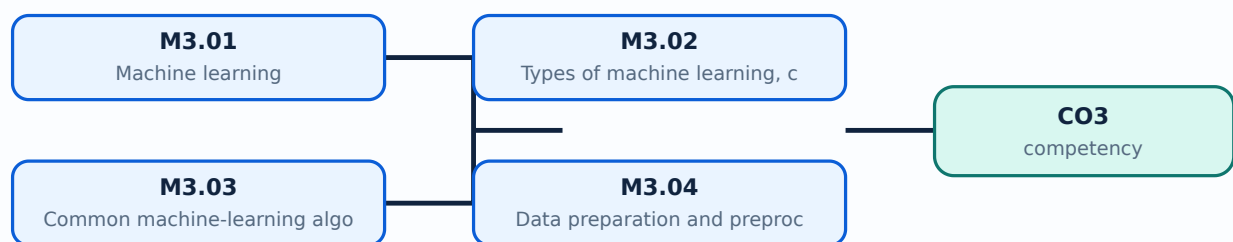
Malayalam support: Random Forest. $\square\square\square\square$ topic $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ exact technical term $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ definition $\square\square\square\square\square\square\square\square$ principle, named parts/steps, application, limitation $\square\square\square\square\square$ $\square\square\square\square\square\square\square$ $\square\square\square\square\square\square\square$. English terminology, symbols, units, diagram labels $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$. Exam answer $\square\square\square\square\square\square\square\square$ concept- $\square\square\square\square$ $\square\square\square\square$ - $\square\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square\square\square$.

Learning goals

- Explain machine learning and supervised/unsupervised learning.
- Describe classification, regression, clustering, association, and named algorithms.
- Prepare data and outline the steps for building a classifier in Python.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

Concept and explanation

Machine learning is a method in which a system learns patterns or parameters from data to make predictions, decisions, or descriptions. A workflow includes problem definition, data collection, preparation, model selection, training, validation, evaluation, deployment, and monitoring. Generalisation to unseen data is more important than memorising training examples.

Malayalam support: Machine learning data- $\square\square\square\square$ pattern/parameters $\square\square\square\square\square\square$ prediction or description $\square\square\square\square\square\square$. Training data- $\square\square\square$ $\square\square\square\square\square$ performance $\square\square\square$, unseen data- $\square\square\square$ generalisation $\square\square\square$ $\square\square\square\square\square\square\square\square$.

Study points

- Define training, validation, test, prediction, and generalisation.
- Separate problem definition from algorithm selection.
- State why data quality and evaluation matter.

Engineering / workplace application: Machine learning supports predictive maintenance, classification, forecasting, recommendation, and anomaly detection.

Illustrative example: A classifier is evaluated on held-out examples that were not used to fit its parameters.

Common mistake: A model that memorises training records can appear accurate but fail on new data.

Handbook treatment

M3.01 — Machine learning (2 hours) is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope M3.01 — Machine learning (2 hours) using the official terminology retained above.
- Organise the important elements of M3.01 — Machine learning (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.01 — Machine learning (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on M3.01 — Machine learning (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.01 — Machine learning (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, M3.01 — Machine learning (2 hours) is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.

Required: construction, working, selection, performance, or diagnosis as requested.

Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.

Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on M3.01 — Machine learning (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.01 — Machine learning (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.01 — Machine learning (2 hours)?
- How would you distinguish M3.01 — Machine learning (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.01 — Machine learning (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.01 — Machine learning (2 hours) incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: M3.01 — Machine learning (2 hours) താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term ഉപയോഗിക്കുക. definition, principle, named parts/steps, application, limitation എന്നിവ ഉൾപ്പെടെയുള്ള English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer concept-ബേസ്ഡ് രീതിയിൽ ഉപയോഗിക്കുക.

– M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours)

Concept and explanation

Supervised learning uses labelled inputs. Classification predicts discrete classes, while regression predicts a continuous value. Unsupervised learning works without target labels; clustering groups similar samples and association discovers co-occurring items or rules. The task, target variable, data type, and evaluation measure determine whether a method is appropriate. Data leakage must be avoided when preparing training and testing sets.

Malayalam support: Classification category predict ഫലപ്രദമാണ്; regression continuous value predict തുല്യമാണ്; clustering similar samples group തുല്യമാണ്; association co-occurrence rules തുല്യമാണ്. Label and task തുല്യമാണ് method തുല്യമാണ്.

Study points

- Compare supervised and unsupervised learning.
- Differentiate classification, regression, clustering, and association.
- Explain why leakage between training and test data is a problem.

Engineering / workplace application: Quality classification, customer grouping, demand regression, and product association are common examples.

Illustrative example: Predicting pass/fail is classification; predicting marks is regression; grouping customers without labels is clustering.

Common mistake: Using the target outcome while preparing a feature leaks the answer into the model and gives misleading evaluation.

Handbook treatment

M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours) is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours) using the official terminology retained above.
- Organise the important elements of M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours) is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.

Required: construction, working, selection, performance, or diagnosis as requested.

Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.

Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours)?
- How would you distinguish M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours) incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: M3.02 — Types of machine learning, classification, regression, clustering and association (4 hours) താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term ഉപയോഗിച്ച് definition, principle, named parts/steps, application, limitation തുടങ്ങിയവയെ സംബന്ധിച്ച് വിശദീകരിക്കുക. English terminology, symbols, units, diagram labels ഉപയോഗിച്ച് വിശദീകരിക്കുക. Exam answer രീതിയിൽ concept-ങ്ങൾ സംബന്ധിച്ച് വിശദീകരിക്കുക.

– M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours)

Concept and explanation

Linear regression models a continuous output from input features. K-Nearest Neighbours (KNN) predicts from nearby training samples and is sensitive to scale and the chosen k. K Means partitions samples into a chosen number of clusters by iteratively updating centroids. Support Vector Machines (SVM) seek a separating boundary with a margin and can use kernels for non-linear separation. Algorithm choice depends on data size, feature scale, interpretability, noise, and task.

Malayalam support: Linear regression continuous output model $y = a + bx$. KNN nearby samples $\frac{1}{n} \sum_{i=1}^n x_i$; K Means centroid $\frac{1}{k} \sum_{i=1}^k x_i$ clusters $\{x_1, x_2, \dots, x_k\}$; SVM separating margin $\frac{1}{\sqrt{a^2 + b^2}}$. Feature scaling and task $\frac{1}{\sqrt{a^2 + b^2}}$ algorithm $\frac{1}{\sqrt{a^2 + b^2}}$.

Study points

- State the basic idea and primary use of each named algorithm.
- Mention scaling sensitivity and parameter choices.
- Distinguish regression from classification use.

Engineering / workplace application: Engineers compare baseline and advanced models using appropriate metrics rather than selecting by name alone.

Illustrative example: Before KNN, features with very different units may be scaled so one large-valued feature does not dominate distance.

Common mistake: K Means cluster labels are not automatically meaningful classes; they need interpretation.

Handbook treatment

M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours) using the official terminology retained above.
- Organise the important elements of M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours)?
- How would you distinguish M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours) incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M3.03 — Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM (6 hours)  topic

1. **Definition:** exact technical term (e.g., "photosynthesis").
 2. **Principle:** principle, named parts/steps, application, limitation (e.g., "photosynthesis converts light energy into chemical energy").
 3. **Terminology:** English terminology, symbols, units, diagram labels (e.g., "chlorophyll", "ATP", "glucose").
 4. **Exam answer:** concept-based answer (e.g., "photosynthesis is the process by which plants convert light energy into chemical energy").

– M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours)

Concept and explanation

Data preparation checks types, missing values, duplicates, outliers, units, labels, and feature meaning. NumPy arrays support numerical operations and vectorised preprocessing. Binarization converts values using a threshold; mean removal centres a feature around zero; scaling adjusts feature ranges; normalization can transform a vector to a common magnitude or distribution depending on the method. Preprocessing parameters should be learned from training data and applied consistently to validation and test data.

Malayalam support: Data preprocessing- $\square$ missing values, outliers, units, labels $\square\square\square\square\square\square\square\square\square\square\square\square$. Binarization threshold $\square\square\square\square\square\square\square\square\square$; mean removal centre $\square\square\square\square\square\square$; scaling range $\square\square\square\square\square\square$; normalization common scale $\square\square\square\square$.

Study points

- Explain the purpose of each listed preprocessing technique.
- Use training data to fit preprocessing parameters.
- Check that preprocessing does not change label meaning or leak test information.

Engineering / workplace application: Preprocessing is essential for distance-based algorithms, numerical stability, and reliable model comparison.

Illustrative example: Compute the training mean and scale, transform training data, and reuse those same values for new samples.

Common mistake: Fitting a scaler on the complete dataset before splitting can leak test-distribution information.

Handbook treatment

M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours) using the official terminology retained above.
- Organise the important elements of M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours)?
- How would you distinguish M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.

- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.04 — Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization (4 hours)
topic
exact technical term
definition
principle, named parts/steps, application, limitation
English terminology, symbols, units, diagram labels
Exam answer
concept-
-
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– M3.05 — Steps for building a classifier in Python (2 hours)

Concept and explanation

A classifier workflow is: define the target and classes; collect and inspect data; clean and prepare features; split into training and test sets; choose a baseline algorithm; fit the model; predict test samples; evaluate with suitable metrics; inspect errors; and save or deploy the validated pipeline. The process should be reproducible with fixed configuration, recorded data assumptions, and versioned code.

Malayalam support: Classifier build ഫ്ലോചാർട്ട് target define, data inspect, preprocess, train/test split, model fit, predict, evaluate, error inspect, deploy ഫ്ലോചാർട്ട് sequence ഫ്ലോചാർട്ട്. Reproducibility-ഫ്ലോചാർട്ട് assumptions ഫ്ലോചാർട്ട്.

Study points

- Write the classifier sequence in correct order.
- Explain why a test set is held back.
- Include error analysis and monitoring after evaluation.

Engineering / workplace application: A Python classifier can support fault categories, document classification, or image labels when the data and evaluation are appropriate.

Illustrative example: A workflow can train Naïve Bayes, SVM, Decision Tree, or Random Forest models and compare validation metrics before selecting one.

Common mistake: Reporting only training accuracy is not evidence of useful generalisation.

Handbook treatment

M3.05 — Steps for building a classifier in Python (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.05 — Steps for building a classifier in Python (2 hours) using the official terminology retained above.
- Organise the important elements of M3.05 — Steps for building a classifier in Python (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.05 — Steps for building a classifier in Python (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Machine Learning with Python.
- Write an examination answer on M3.05 — Steps for building a classifier in Python (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.05 — Steps for building a classifier in Python (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.05 — Steps for building a classifier in Python (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.05 — Steps for building a classifier in Python (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.05 — Steps for building a classifier in Python (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.05 — Steps for building a classifier in Python (2 hours)?
- How would you distinguish M3.05 — Steps for building a classifier in Python (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.05 — Steps for building a classifier in Python (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.05 — Steps for building a classifier in Python (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.05 — Steps for building a classifier in Python (2 hours)

topic exact technical term definition principle, named parts/steps, application, limitation English terminology, symbols, units, diagram labels Exam answer concept- process- outcome.

Module summary: Machine learning is a disciplined data workflow: choose the task, prepare data without leakage, fit, evaluate, and monitor.

OFFICIAL MODULE 4

AI with Python in Gaming

Games provide a compact environment for search, decision-making, state representation, and evaluation. The goal is to connect AI concepts to a bot that chooses legal and useful moves.

Hours: 10

CO4 · Bloom level: Applying

Handbook study routine: Complete Module 4 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 4

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

[View extracted official source transcript](#)

Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play games, Bot to play last coin standing game, Bot to play tic tac toe

Point-by-point study guide

Exact source point

Syllabus text: Search Algorithms- Combinational Search, Minimax Algorithm.
Building bots to play

How to understand and study it

This point is an official syllabus requirement: “Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play”. Study the input, logic or algorithm, output, and one test condition. Write the steps in order and identify the common error that changes the result. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Practise one small input-output example and test at least one boundary or fault condition.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: syllabus point ഫലപ്രദമായി Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play ഫലപ്രദ technical terms English-ഫലപ്രദമായി. ഫലപ്രദ definition ഫലപ്രദമായി principle ഫലപ്രദമായി; ഫലപ്രദ term-ഫലപ്രദ function, difference, application ഫലപ്രദ ഫലപ്രദമായി ഫലപ്രദമായി. Diagram, sequence, formula, unit ഫലപ്രദമായി ഫലപ്രദമായി ഫലപ്രദമായി revision ഫലപ്രദമായി.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play using the official terminology retained above.
- Organise the important elements of Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play into a logical sequence, classification, structure, or calculation.
- Connect Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play with one engineering decision, operation, measurement, or safety requirement in the context of AI with Python in Gaming.
- Write an examination answer on Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play?
- How would you distinguish Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play from a closely related idea?
- Where would an engineer or technician encounter Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play, and what must be checked before using it?
- What common mistake would make an answer or operation involving Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: Search Algorithms- Combinational Search, Minimax Algorithm. Building bots to play ടാപ്പിക് ടാപ്പിക് exact technical term ടാപ്പിക്. definition principle, named parts/steps, application, limitation ടാപ്പിക്. English terminology, symbols, units, diagram labels ടാപ്പിക്. Exam answer concept-ടാപ്പിക്-ടാപ്പിക് ടാപ്പിക്.

– Official syllabus point 2: games, Bot to play last coin standing game, Bot to play tic tac toe

Exact source point

Syllabus text: games, Bot to play last coin standing game, Bot to play tic tac toe

How to understand and study it

This point is an official syllabus requirement: “games, Bot to play last coin standing game, Bot to play tic tac toe”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് Bot to play last coin standing game, Bot to play tic tac toe ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

games, Bot to play last coin standing game, Bot to play tic tac toe is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope games, Bot to play last coin standing game, Bot to play tic tac toe using the official terminology retained above.
- Organise the important elements of games, Bot to play last coin standing game, Bot to play tic tac toe into a logical sequence, classification, structure, or calculation.
- Connect games, Bot to play last coin standing game, Bot to play tic tac toe with one engineering decision, operation, measurement, or safety requirement in the context of AI with Python in Gaming.
- Write an examination answer on games, Bot to play last coin standing game, Bot to play tic tac toe with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading games, Bot to play last coin standing game, Bot to play tic tac toe. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of games, Bot to play last coin standing game, Bot to play tic tac toe is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on games, Bot to play last coin standing game, Bot to play tic tac toe, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is games, Bot to play last coin standing game, Bot to play tic tac toe, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining games, Bot to play last coin standing game, Bot to play tic tac toe?
- How would you distinguish games, Bot to play last coin standing game, Bot to play tic tac toe from a closely related idea?
- Where would an engineer or technician encounter games, Bot to play last coin standing game, Bot to play tic tac toe, and what must be checked before using it?
- What common mistake would make an answer or operation involving games, Bot to play last coin standing game, Bot to play tic tac toe incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: games, Bot to play last coin standing game, Bot to play tic tac toe **topic** **exact technical term** **definition** **principle, named parts/steps, application, limitation** **English terminology, symbols, units, diagram labels** **Exam answer** **concept-** **-** .

Learning goals

- Describe search algorithms and combinational search.
- Explain Minimax for adversarial games.
- Outline bots for last-coin-standing and Tic Tac Toe.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

– M4.01 — Search algorithms (2 hours)

Concept and explanation

A search algorithm explores possible states or actions to find a path, decision, or goal. A game state includes the current position, legal moves, player turn, and terminal condition. Search may be uninformed or guided by a heuristic. Correct state representation, termination, duplicate-state handling, and computational cost are essential.

Malayalam support: Search algorithm possible states/moves explore ഏതു ഏതു goal decision ഏതു ഏതു. State representation, legal moves, terminal condition, heuristic, computation cost ഏതു ഏതു.

Study points

- Define state, action, goal, and terminal condition.
- Explain why branching factor affects computation.
- Use a small state-space example for tracing.

Engineering / workplace application: Game bots, route planning, puzzle solving, and scheduling use search concepts.

Illustrative example: A bot can represent Tic Tac Toe as a board state and recursively explore legal moves until win, draw, or loss.

Common mistake: A search that does not detect terminal states may continue after the game is already over.

Handbook treatment

M4.01 — Search algorithms (2 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M4.01 — Search algorithms (2 hours) using the official terminology retained above.
- Organise the important elements of M4.01 — Search algorithms (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.01 — Search algorithms (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of AI with Python in Gaming.
- Write an examination answer on M4.01 — Search algorithms (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.01 — Search algorithms (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M4.01 — Search algorithms (2 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M4.01 — Search algorithms (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.01 — Search algorithms (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.01 — Search algorithms (2 hours)?
- How would you distinguish M4.01 — Search algorithms (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.01 — Search algorithms (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.01 — Search algorithms (2 hours) incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M4.01 — Search algorithms (2 hours) താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term ഉപയോഗിക്കുക. definition, principle, named parts/steps, application, limitation എന്നിവ ഉൾക്കൊള്ളിക്കുക. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer concept-ബേസ്ഡ് ആയിരിക്കണം.

Concept and explanation

Combinational search explores combinations of choices. In a two-player zero-sum game, Minimax models one player maximising utility while the opponent minimises it. The algorithm expands a game tree to terminal or depth-limited states and backs up scores. A heuristic evaluation estimates non-terminal positions; alpha-beta pruning can reduce explored branches without changing the Minimax result when applied correctly.

Malayalam support: Minimax-ൽ one player score maximise ചെയ്യുന്നു, opponent minimise ചെയ്യുന്നു. Game tree terminal/depth-limited states-ൽ expand ചെയ്ത് scores back up ചെയ്യുന്നു. Heuristic non-terminal state evaluate ചെയ്യുന്നു.

Study points

- Explain MAX and MIN levels.
- Trace a small game tree and select the backed-up move.
- State the purpose of depth limits and heuristic evaluation.

Engineering / workplace application: Adversarial search is used in board games and decision problems where another agent acts against the bot.

Illustrative example: At a MAX node the bot chooses the largest child value; at a MIN node the opponent is assumed to choose the smallest child value.

Common mistake: Choosing the move with the best immediate score without considering the opponent is not Minimax.

Handbook treatment

M4.02 — Combinational search and Minimax algorithm (3 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M4.02 — Combinational search and Minimax algorithm (3 hours) using the official terminology retained above.
- Organise the important elements of M4.02 — Combinational search and Minimax algorithm (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.02 — Combinational search and Minimax algorithm (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of AI with Python in Gaming.
- Write an examination answer on M4.02 — Combinational search and Minimax algorithm (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.02 — Combinational search and Minimax algorithm (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M4.02 — Combinational search and Minimax algorithm (3 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M4.02 — Combinational search and Minimax algorithm (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.02 — Combinational search and Minimax algorithm (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.02 — Combinational search and Minimax algorithm (3 hours)?
- How would you distinguish M4.02 — Combinational search and Minimax algorithm (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.02 — Combinational search and Minimax algorithm (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.02 — Combinational search and Minimax algorithm (3 hours) incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M4.02 — Combinational search and Minimax algorithm (3 hours)
topic
exact technical term
definition
principle, named parts/steps, application, limitation
English terminology, symbols, units, diagram labels
Exam answer
concept-
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– M4.03 — Building bots to play games (1 hours)

Concept and explanation

A game bot needs a state representation, legal-move generator, terminal-test function, decision algorithm, evaluation function, action executor, and test cases. The bot should handle invalid moves, draw conditions, and deterministic or controlled randomness. Separating game rules from search logic makes the program easier to debug and reuse.

Malayalam support: Game bot-`state`, legal moves, terminal test, decision algorithm, evaluation, action execution `function`. Rules and search logic `function` debugging `function`.

Study points

- List the components of a game bot.
- Keep rules and decision logic separate.
- Test win, loss, draw, invalid, and edge cases.

Engineering / workplace application: Small games are useful for demonstrating AI decisions without needing large datasets.

Illustrative example: A reusable `GameState` object can provide legal moves and terminal status to a Minimax function.

Common mistake: A bot that only chooses a move but does not validate legality can corrupt the game state.

Handbook treatment

M4.03 — Building bots to play games (1 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.03 — Building bots to play games (1 hours) using the official terminology retained above.
- Organise the important elements of M4.03 — Building bots to play games (1 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.03 — Building bots to play games (1 hours) with one engineering decision, operation, measurement, or safety requirement in the context of AI with Python in Gaming.
- Write an examination answer on M4.03 — Building bots to play games (1 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.03 — Building bots to play games (1 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.03 — Building bots to play games (1 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.03 — Building bots to play games (1 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.03 — Building bots to play games (1 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.03 — Building bots to play games (1 hours)?
- How would you distinguish M4.03 — Building bots to play games (1 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.03 — Building bots to play games (1 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.03 — Building bots to play games (1 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.03 — Building bots to play games (1 hours) താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term ഉപയോഗിക്കുക. definition, principle, named parts/steps, application, limitation എന്നിവ ഉൾപ്പെടെയുള്ള English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer concept-ബോക്സ് ഉപയോഗിക്കുക.

– M4.04 — Bot to play last coin standing game (2 hours)

Concept and explanation

In a last-coin-standing game, players remove an allowed number of coins until a terminal condition is reached. The bot represents the remaining coins and player turn, generates legal removals, evaluates winning or losing states, and selects a move using a rule or search. The exact winning condition and allowed removal count must be stated before coding because they change the strategy.

Malayalam support: Last coin game- $\square$ remaining coins, player turn, allowed removal, win condition $\square\square\square\square\square$ state $\square\square\square$ $\square\square\square\square\square\square\square\square\square\square$. Rules $\square\square\square\square\square\square\square\square\square\square\square\square\square\square\square\square$ strategy $\square\square\square\square$.

Study points

- Define the game rule before implementing.
- Generate only legal moves.
- Trace a small state tree and test terminal cases.

Engineering / workplace application: This game demonstrates state-space search, terminal evaluation, and deterministic decision logic.

Illustrative example: If a player may remove one or two coins, a bot can evaluate each legal move recursively and select the move leading to a winning state.

Common mistake: Assuming the last coin rule or allowed move set without documenting it produces a different game.

Handbook treatment

M4.04 — Bot to play last coin standing game (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.04 — Bot to play last coin standing game (2 hours) using the official terminology retained above.
- Organise the important elements of M4.04 — Bot to play last coin standing game (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.04 — Bot to play last coin standing game (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of AI with Python in Gaming.
- Write an examination answer on M4.04 — Bot to play last coin standing game (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.04 — Bot to play last coin standing game (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.04 — Bot to play last coin standing game (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.04 — Bot to play last coin standing game (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.04 — Bot to play last coin standing game (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.04 — Bot to play last coin standing game (2 hours)?
- How would you distinguish M4.04 — Bot to play last coin standing game (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.04 — Bot to play last coin standing game (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.04 — Bot to play last coin standing game (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.04 — Bot to play last coin standing game (2 hours)

topic exact technical term definition principle, named parts/steps, application, limitation English terminology, symbols, units, diagram labels Exam answer concept- -

– M4.05 — Bot to play Tic Tac Toe (2 hours)

Concept and explanation

Tic Tac Toe is a finite two-player game with a 3 by 3 board. A bot checks legal empty cells, detects rows, columns, and diagonals, evaluates wins, losses, and draws, and can use Minimax to choose an optimal move for a small complete game tree. The implementation must reject occupied-cell moves and stop immediately at a terminal board.

Malayalam support: Tic Tac Toe board 3×3 □□□. Empty cells, rows/columns/diagonals, win/loss/draw, legal move checking □□□□□□ implement □□□□□□□. Minimax complete small tree explore □□□□□□□.

Study points

- Describe board, legal moves, terminal tests, and evaluation.
- Use Minimax conceptually to choose a move.
- Test immediate win, block, draw, and occupied-cell cases.

Engineering / workplace application: Tic Tac Toe is a clear laboratory example for adversarial search and game-state evaluation.

Illustrative example: A strong bot first checks for an immediate winning move, then blocks an immediate loss, and otherwise applies its search policy.

Common mistake: Continuing to accept moves after a win or ignoring the opponent threat creates an invalid bot.

Handbook treatment

M4.05 — Bot to play Tic Tac Toe (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.05 — Bot to play Tic Tac Toe (2 hours) using the official terminology retained above.
- Organise the important elements of M4.05 — Bot to play Tic Tac Toe (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.05 — Bot to play Tic Tac Toe (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of AI with Python in Gaming.
- Write an examination answer on M4.05 — Bot to play Tic Tac Toe (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.05 — Bot to play Tic Tac Toe (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.05 — Bot to play Tic Tac Toe (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.05 — Bot to play Tic Tac Toe (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.05 — Bot to play Tic Tac Toe (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.05 — Bot to play Tic Tac Toe (2 hours)?
- How would you distinguish M4.05 — Bot to play Tic Tac Toe (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.05 — Bot to play Tic Tac Toe (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.05 — Bot to play Tic Tac Toe (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.05 — Bot to play Tic Tac Toe (2 hours) താഴെ വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term ഉപയോഗിക്കുക. definition, principle, named parts/steps, application, limitation എന്നിവ ഉൾപ്പെടെ. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer concept-ബോക്സ് ഉപയോഗിക്കുക.

Module summary: Game AI makes search concrete: represent state, generate legal actions, evaluate consequences, and test every terminal condition.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

[Click here to view its labelled learning-map diagram.](#)

Module 2: Python

[Click here to view its labelled learning-map diagram.](#)

Module 3: Machine Learning

[Click here to view its labelled learning-map diagram.](#)

Module 4: AI with Python in

[Open the module to view its labelled learning-map diagram.](#)

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied syllabus does not provide a separate official self-learning activity list for this theory course. Use the topic procedures, worked examples, and module questions in this lesson for structured study.

Practical requirements / equipment

- The supplied syllabus does not prescribe separate practical equipment for this theory course.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Source anomaly: The supplied Course Outcomes table lists Series Test as 2 hours, while the course outline separately lists Series Test I and Series Test II as 1 hour within Modules 2 and 4; both official entries are preserved.
- No separate CIE/ESE mark distribution is stated in the supplied PDF.

Module-wise Questions and Answers

module-1 — Fundamental Understanding of Artificial Intelligence

1. Explain Artificial Intelligence and the necessity of learning AI. [3 marks]

Artificial Intelligence is the design of systems that perform tasks requiring capabilities such as perception, reasoning, planning, language understanding, learning, or decision-making. Learning AI is necessary because data, automation, intelligent assistance, robotics, recommendation, and decision-support are present across engineering and society. A responsible AI learner also studies data quality, bias, privacy, safety, and limits of automated decisions.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Different types of learning. [3 marks]

At an introductory level, supervised learning uses labelled examples to learn a mapping for prediction or classification. Unsupervised learning searches for structure in unlabelled data, such as clusters or associations. Reinforcement learning learns actions from interaction, rewards, and penalties. The choice of learning type depends on available data, target task, feedback, and evaluation method.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Fields of AI. [3 marks]

AI includes fields such as machine learning, natural-language processing, computer vision, robotics, expert systems, search and planning, speech processing, knowledge representation, and game AI. Fields often overlap: a robot may use vision, planning, learning, and control together. The field should be selected from the problem capability required, not from a fashionable label.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Applications of AI. [3 marks]

Applications include image and speech recognition, recommendation, medical and industrial decision support, autonomous systems, fraud detection, predictive maintenance, chat interfaces, games, and optimisation. Each application has a task, data, model or rule system, evaluation metric, deployment environment, and risk profile. A useful application must be validated against real requirements and monitored after deployment.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Python Programming for AI

5. Explain Role, features and installation of Python. [3 marks]

Python is used in AI because its syntax supports rapid experimentation and its ecosystem includes numerical, data, visualisation, machine-learning, and automation libraries. Important features include interpreted execution, dynamic typing, indentation-based blocks, portability,

and extensive modules. Installation requires a supported Python interpreter and a controlled package environment; version and dependency records improve reproducibility.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Python basics: data types, conditionals, loops, control statements and strings. [3 marks]

Python basics include numeric, Boolean, string, list, tuple, set, dictionary, and None values; variables refer to objects and expressions produce values. Conditional statements select paths, loops repeat work, and break/continue control iteration. Strings are immutable sequences supporting indexing, slicing, methods, formatting, and conversion. Correct type conversion and boundary testing are essential when preparing input for an AI program.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain List and dictionary manipulations, functions, modules and packages. [3 marks]

Lists store ordered mutable data; dictionaries store key-value records; functions package reusable logic; modules and packages organise code and dependencies. AI programs commonly represent samples as lists, records as dictionaries, and preprocessing or prediction steps as functions. Clear interfaces, return values, descriptive names, and small tests make experimentation reliable.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Object-oriented programming in Python and regular expressions. [3 marks]

Object-oriented programming groups data and behaviour into classes and objects. Encapsulation, constructors, methods, inheritance, and polymorphism can organise reusable AI components, although simple functions may be sufficient for small tasks. Regular expressions describe text patterns for searching, validation, extraction, and substitution. Exception handling with try/except prevents expected input or file errors from crashing a complete application and should not hide unknown failures.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Machine Learning with Python

9. Explain Machine learning. [3 marks]

Machine learning is a method in which a system learns patterns or parameters from data to make predictions, decisions, or descriptions. A workflow includes problem definition, data collection, preparation, model selection, training, validation, evaluation, deployment, and monitoring. Generalisation to unseen data is more important than memorising training examples.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Types of machine learning, classification, regression, clustering and association. [3 marks]

Supervised learning uses labelled inputs. Classification predicts discrete classes, while regression predicts a continuous value. Unsupervised learning works without target labels; clustering groups similar samples and association discovers co-occurring items or rules. The task, target variable, data type, and evaluation measure determine whether a method is appropriate. Data leakage must be avoided when preparing training and testing sets.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM. [3 marks]

Linear regression models a continuous output from input features. K-Nearest Neighbours (KNN) predicts from nearby training samples and is sensitive to scale and the chosen k. K Means partitions samples into a chosen number of clusters by iteratively updating centroids. Support Vector Machines (SVM) seek a separating boundary with a margin and can use kernels for non-linear separation. Algorithm choice depends on data size, feature scale, interpretability, noise, and task.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization. [3 marks]

Data preparation checks types, missing values, duplicates, outliers, units, labels, and feature meaning. NumPy arrays support numerical operations and vectorised preprocessing.

Binarization converts values using a threshold; mean removal centres a feature around zero; scaling adjusts feature ranges; normalization can transform a vector to a common magnitude or distribution depending on the method. Preprocessing parameters should be learned from training data and applied consistently to validation and test data.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — AI with Python in Gaming

13. Explain Search algorithms. [3 marks]

A search algorithm explores possible states or actions to find a path, decision, or goal. A game state includes the current position, legal moves, player turn, and terminal condition. Search may be uninformed or guided by a heuristic. Correct state representation, termination, duplicate-state handling, and computational cost are essential.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Combinational search and Minimax algorithm. [3 marks]

Combinational search explores combinations of choices. In a two-player zero-sum game, Minimax models one player maximising utility while the opponent minimises it. The algorithm expands a game tree to terminal or depth-limited states and backs up scores. A heuristic evaluation estimates non-terminal positions; alpha-beta pruning can reduce explored branches without changing the Minimax result when applied correctly.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Building bots to play games. [3 marks]

A game bot needs a state representation, legal-move generator, terminal-test function, decision algorithm, evaluation function, action executor, and test cases. The bot should handle invalid moves, draw conditions, and deterministic or controlled randomness. Separating game rules from search logic makes the program easier to debug and reuse.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Bot to play last coin standing game. [3 marks]

In a last-coin-standing game, players remove an allowed number of coins until a terminal condition is reached. The bot represents the remaining coins and player turn, generates legal removals, evaluates winning or losing states, and selects a move using a rule or search. The exact winning condition and allowed removal count must be stated before coding because they change the strategy.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. Define or explain *Artificial Intelligence and the necessity of learning AI* with one relevant application. (5 marks)
2. Define or explain *Different types of learning* with one relevant application. (5 marks)
3. Define or explain *Role, features and installation of Python* with one relevant application. (5 marks)
4. Define or explain *Python basics: data types, conditionals, loops, control statements and strings* with one relevant application. (5 marks)
5. Define or explain *Machine learning* with one relevant application. (5 marks)
6. Define or explain *Types of machine learning, classification, regression, clustering and association* with one relevant application. (5 marks)

7. **7.** Define or explain *Search algorithms* with one relevant application. (5 marks)

8. **8.** Define or explain *Combinational search and Minimax algorithm* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Fundamental Understanding of Artificial Intelligence:** AI is a broad problem-solving field; learning type, field, application, data, and evaluation must be connected.
- **Python Programming for AI:** Python fundamentals become useful AI skills when code is modular, validated, reproducible, and connected to data.
- **Machine Learning with Python:** Machine learning is a disciplined data workflow: choose the task, prepare data without leakage, fit, evaluate, and monitor.
- **AI with Python in Gaming:** Game AI makes search concrete: represent state, generate legal actions, evaluate consequences, and test every terminal condition.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Artificial Intelligence and the necessity of learning AI	Artificial Intelligence is the design of systems that perform tasks requiring capabilities such as perception, reasoning, planning, language understanding, learning, or decision-making. Learning AI is necessary because data, automation, intelligent assistance,
Different types of learning	At an introductory level, supervised learning uses labelled examples to learn a mapping for prediction or classification. Unsupervised learning searches for structure in unlabelled data, such as clusters or associations. Reinforcement learning learns actions f
Fields of AI	AI includes fields such as machine learning, natural-language processing, computer vision, robotics, expert systems, search and planning, speech processing, knowledge representation, and game AI. Fields often overlap: a robot may use vision, planning, learning
Applications of AI	Applications include image and speech recognition, recommendation, medical and industrial decision support, autonomous systems, fraud detection, predictive maintenance, chat interfaces, games, and optimisation. Each application has a task, data, model or rule
Role, features and installation of Python	Python is used in AI because its syntax supports rapid experimentation and its ecosystem includes numerical, data, visualisation, machine-learning, and automation libraries. Important features include interpreted execution, dynamic typing, indentation-based bl
Python basics: data types, conditionals, loops, control statements and strings	Python basics include numeric, Boolean, string, list, tuple, set, dictionary, and None values; variables refer to objects and expressions produce values. Conditional statements select paths, loops repeat work, and break/continue control iteration. Strings are
List and dictionary manipulations, functions, modules and packages	Lists store ordered mutable data; dictionaries store key-value records; functions package reusable logic; modules and packages organise code and dependencies. AI programs commonly represent samples as lists, records as dictionaries, and preprocessing or predic
Object-oriented programming in Python and regular expressions	Object-oriented programming groups data and behaviour into classes and objects. Encapsulation, constructors, methods, inheritance, and polymorphism can organise reusable AI components, although simple functions may be sufficient for small tasks. Regular expres
Machine learning	Machine learning is a method in which a system learns patterns or parameters from data to make predictions, decisions, or descriptions. A workflow includes problem definition, data collection, preparation, model selection, training, validation, evaluation, dep
Types of machine learning, classification, regression, clustering and association	Supervised learning uses labelled inputs. Classification predicts discrete classes, while regression predicts a continuous value. Unsupervised learning works without target labels; clustering groups similar samples and association discovers co-occurring items
Common machine-learning algorithms: Linear Regression, KNN, K Means, Regression and SVM	Linear regression models a continuous output from input features. K-Nearest Neighbours (KNN) predicts from nearby training samples and is sensitive to scale and the chosen k. K Means partitions samples into a chosen number of clusters by iteratively updating c

Term	Short meaning
Data preparation and preprocessing using NumPy: binarization, mean removal, scaling and normalization	Data preparation checks types, missing values, duplicates, outliers, units, labels, and feature meaning. NumPy arrays support numerical operations and vectorised preprocessing. Binarization converts values using a threshold; mean removal centres a feature around
Steps for building a classifier in Python	A classifier workflow is: define the target and classes; collect and inspect data; clean and prepare features; split into training and test sets; choose a baseline algorithm; fit the model; predict test samples; evaluate with suitable metrics; inspect errors;
Search algorithms	A search algorithm explores possible states or actions to find a path, decision, or goal. A game state includes the current position, legal moves, player turn, and terminal condition. Search may be uninformed or guided by a heuristic. Correct state representation
Combinational search and Minimax algorithm	Combinational search explores combinations of choices. In a two-player zero-sum game, Minimax models one player maximising utility while the opponent minimises it. The algorithm expands a game tree to terminal or depth-limited states and backs up scores. A heuristic
Building bots to play games	A game bot needs a state representation, legal-move generator, terminal-test function, decision algorithm, evaluation function, action executor, and test cases. The bot should handle invalid moves, draw conditions, and deterministic or controlled randomness. See
Bot to play last coin standing game	In a last-coin-standing game, players remove an allowed number of coins until a terminal condition is reached. The bot represents the remaining coins and player turn, generates legal removals, evaluates winning or losing states, and selects a move using a rule
Bot to play Tic Tac Toe	Tic Tac Toe is a finite two-player game with a 3 by 3 board. A bot checks legal empty cells, detects rows, columns, and diagonals, evaluates wins, losses, and draws, and can use Minimax to choose an optimal move for a small complete game tree. The implementation

Official References and Resources

References listed in the supplied syllabus

1. Alberto Artasánchez and Prateek Joshi, Artificial Intelligence with Python: Your complete guide to building intelligent apps using Python 3.x and TensorFlow 2, 2nd Edition.
2. Nageswara Rao, Core Python Programming.
3. Yashvant Kanetkar, Let us Python.

Official learning resources listed

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